Journal of Neural Engineering

PAPER

Transferable non-invasive modal fusion-transformer (NIMFT) for end-to-end hand gesture recognition

To cite this article: Tianxiang Xu *et al* 2024 *J. Neural Eng.* **21** 026034

View the article online for updates and enhancements.

Journal of Neural Engineering

CrossMark

RECEIVED
2 November 2023

REVISED
7 March 2024

ACCEPTED FOR PUBLICATION
2 April 2024

PUBLISHED
9 April 2024

PAPER

Transferable non-invasive modal fusion-transformer (NIMFT) for end-to-end hand gesture recognition

Tianxiang Xu[1,2,4], Kunkun Zhao[1,2,4], Yuxiang Hu[1,2], Liang Li[1,2], Wei Wang[1,2], Fulin Wang[1,3], Yuxuan Zhou[1,2,*] and Jianqing Li[1,2,*]

1 School of Biomedical Engineering and Informatics, Nanjing Medical University, Nanjing 211166, People's Republic of China
2 The Engineering Research Center of Intelligent Theranostics Technology and Instruments, Ministry of Education, School of Biomedical Engineering and Informatics, Nanjing Medical University, Nanjing 211166, People's Republic of China
3 Nanjing PANDA Electronics Equipment Co., Ltd, Nanjing 210033, People's Republic of China
4 These authors contributed equally to this work.
* Authors to whom any correspondence should be addressed.

E-mail: yxzhou@njmu.edu.cn and jqli@njmu.edu.cn

Keywords: hand gesture recognition, multimodal fusion, deep transfer learning, sEMG, acceleration, transformer

Supplementary material for this article is available online

Abstract

Objective. Recent studies have shown that integrating inertial measurement unit (IMU) signals with surface electromyographic (sEMG) can greatly improve hand gesture recognition (HGR) performance in applications such as prosthetic control and rehabilitation training. However, current deep learning models for multimodal HGR encounter difficulties in invasive modal fusion, complex feature extraction from heterogeneous signals, and limited inter-subject model generalization. To address these challenges, this study aims to develop an end-to-end and inter-subject transferable model that utilizes non-invasively fused sEMG and acceleration (ACC) data. *Approach.* The proposed non-invasive modal fusion-transformer (NIMFT) model utilizes 1D-convolutional neural networks-based patch embedding for local information extraction and employs a multi-head cross-attention (MCA) mechanism to non-invasively integrate sEMG and ACC signals, stabilizing the variability induced by sEMG. The proposed architecture undergoes detailed ablation studies after hyperparameter tuning. Transfer learning is employed by fine-tuning a pre-trained model on new subject and a comparative analysis is performed between the fine-tuning and subject-specific model. Additionally, the performance of NIMFT is compared to state-of-the-art fusion models. *Main results.* The NIMFT model achieved recognition accuracies of 93.91%, 91.02%, and 95.56% on the three action sets in the Ninapro DB2 dataset. The proposed embedding method and MCA outperformed the traditional invasive modal fusion transformer by 2.01% (embedding) and 1.23% (fusion), respectively. In comparison to subject-specific models, the fine-tuning model exhibited the highest average accuracy improvement of 2.26%, achieving a final accuracy of 96.13%. Moreover, the NIMFT model demonstrated superiority in terms of accuracy, recall, precision, and F1-score compared to the latest modal fusion models with similar model scale. *Significance.* The NIMFT is a novel end-to-end HGR model, utilizes a non-invasive MCA mechanism to integrate long-range intermodal information effectively. Compared to recent modal fusion models, it demonstrates superior performance in inter-subject experiments and offers higher training efficiency and accuracy levels through transfer learning than subject-specific approaches.

1. Introduction

With the development of machine learning (ML) and deep neural networks (DNNs), hand gesture recognition (HGR) is widely used and developed as a new human-machine interaction (HMI) technology [1]. HGR with surface electromyography (sEMG) signals is a key method for identifying motion intentions, widely applied in prosthetic control [2], exoskeleton assistance [3], sign language interpretation [4], rehabilitation [5] and so on. The primary goal of sEMG-based HGR research, as reported in the literature, is to create models requiring minimal training while ensuring high accuracy in plug-and-play scenarios [6]. sEMG signals can be classified into sparse sEMG and high-density sEMG (HD-sEMG) based on electrode configurations. HD-sEMG signals possess the ability to capture comprehensive temporal and spatial information of muscle activities from the entire muscle covered area, leading to enhanced accuracy in classification [7, 8]. Geng *et al* [7] proposed an end-to-end convolutional neural networks (CNN) model that achieved 99.6% accuracy using HD-sEMG. However, when tested with sparse multi-channel sEMG, the accuracy dropped to 77.8%. Despite its superior classification accuracy [9, 10], HD-sEMG requires multiple channels for high-speed and real-time sampling through dozens or even hundreds of electrodes. This systematic complexity and computational demand hinder practical applications. Thus, sparse multi-channel sEMG, which uses fewer electrodes, has become the most commonly used wearable system technique for recording muscle activity [11].

Previous research on sparse multi-channel sEMG gesture recognition aimed at improving classification accuracy and efficiency by introducing new features [12, 13] and evaluating existing ones [14, 15], to identify an optimal subset, which were then input ML classifiers like random forest [16], linear discriminant analysis (LDA) [17], and support vector machine (SVM) [18]. Recent studies have demonstrated the superiority of deep learning models, such as CNN [19, 20], recurrent neural networks (RNN) [21, 22], and transformer architecture [23], over conventional approaches in gesture classification. These advanced deep architectures exhibit exceptional proficiency in extracting high-level feature information from multiple representative and hidden layers [24]. Ameur *et al* [21] introduced a hybrid architecture, hybrid bidirectional unidirectional LSTM (HBU-LSTM), combining unidirectional and bidirectional LSTM, achieving accuracies of 89.98% and 73.95% on the LeapGestureDB dataset and the RIT dataset, respectively. Shen *et al* [23] enhanced the vision transformer by adding convolutional layers, achieving accuracies of 83.47% and 84.09% for 200 ms and 300 ms window lengths, respectively, on part of NinaPro DB2. However, the commercial uptake of sEMG-based HGR is hindered by issues like high variance between subjects [25], electrode shift [25, 26], and skin fat ratio [27], leading to lower accuracy and the necessity for extensive subject-specific training. Recent research has shown that combining sEMG signals with acceleration (ACC) data enhances the performance of sEMG-based HGR, outperforming the use of sEMG signals alone [10]. sEMG provides richer intrinsic muscle kinematic information [28], while ACC offers a higher signal-to-noise ratio and captures both limb position and dynamic information [29].

However, feature engineering for HGR using ACC signals is still underdeveloped, lacking consensus on the best ACC features. Some studies adopt similar feature extraction methods as used for sEMG signals [30], while others rely on basic features such as mean, maximum, and minimum values [31]. An end-to-end deep learning model that fuses sEMG and ACC modalities enables automatic feature learning without redundant manual crafting. This approach may effectively leverage the complementary information from sEMG and ACC signals, leading to improved HGR performance.

The fusion of sEMG and ACC modalities in prior studies often involved direct fusion methods such as summation or concatenation, known as the invasive method. However, this approach lacks multi-level intermodal interactive knowledge as it performs fusion solely in either the low-level data space or high-level decision space [30]. Moreover, the inherent heterogeneity of these signals may result in disruptions of the original signal characteristics during invasive fusion. To address these limitations, some researchers suggest using correlated coupling between sEMG and ACC to indirectly merge these signals. Zhou *et al* [31] introduced a non-invasive dual attention (NIDA) module to seamlessly integrate sEMG and ACC data in both temporal and spatial dimensions, achieving accurate prediction of four types of lower limb gait movements. However, evaluating optimal input channels separately for temporal and spatial attention modules could be burdensome with many sEMG and ACC channels. Furthermore, existing research has not yet explored if non-invasive fusion methods can improve fusion performance in the more complex HGR.

Another significant challenge in DL based HGR is the issue of model generalization across subjects and days because of variations in sEMG and ACC signals from different individuals and detection sites. In particular, for models with a large parameter count, acquiring ample patient-specific data and implementing subject-specific training proves time-consuming and impractical in real-world scenarios. Transfer learning (TL) has become a key approach to tackle inter-subject HGR challenges [32]. TL involves pre-training a model on multiple subjects and fine-tuning the pretrained model on new subjects. It enables the application of a well-established generalized model to new subjects [32–34]. As highlighted in [32], transfer learning may offer several advantages over subject-specific methods, including (1) a higher initial performance level with faster convergence, (2) improved overall asymptotic performance, (3) reduced data requirements for new subjects, and (4) the ability to achieve satisfactory performance by selectively retraining specific layers. Thus, when evaluating the transferability of a specific HGR model, it becomes imperative to compare models tailored to specific conditions utilizing transfer learning in terms of these four aspects.

In summary, previous studies have demonstrated superior performance in HGR by incorporating both sEMG and ACC signals compared to using sEMG alone. However, most of these studies often required extensive feature extraction, lacked inter-subject experiments, and did not evaluate the transferability of the models. Notably, existing end-to-end sEMG-ACC fusion deep learning models, like those in [31, 35], have not performed inter-subject tests on public datasets for benchmarking against established model. The absence of inter-subject evaluation and transferability assessment reduces the real-world applicability of these models. The main contributions of this paper can be summarized as follows:

(1) We proposed a novel end-to-end, non-invasive modal fusion transformer (NIMFT) that combines sEMG and ACC data to tackle the challenges of complex feature engineering and invasive fusion in existing HGR research. By incorporating a 1D-CNN embedding layer and multi-head cross-attention mechanism (MCA), our model achieved an impressive inter-subject recognition accuracy of 91.92% ± 3.91% on the Ninapro DB2 dataset, which comprises 49 gestures involving functional hand and wrist movements as well as finger force patterns.

(2) We validated the practicality of our model through comprehensive transferability tests. We evaluated the performance of fine-tuning models and subject-specific models based on the four above mentioned metrics of transfer learning. The results demonstrate that our transferable models outperform subject-specific models, exhibiting superior training efficiency and achieving higher accuracy levels.

(3) We conducted inter-subject experiments using a universal model trained on data from 40 subjects in DB2 and evaluated on unseen data. Furthermore, we compared our method with state-of-the-art models on public datasets, demonstrating its superior inter-subject recognition performance. These results underscore the superior performance of our approach in cross-subject HGR tasks.

The rest of this paper is organized as follows. Section 2 offers a comprehensive introduction to the proposed NIMFT architecture. Detailed setup and analysis of the experiments are followed in section 3. In section 4, a thorough discussion of the experimental results and future work is provided. Finally, we conclude the research findings and their significance in section 5.

2. Non-invasive modal fusion transformer framework

NIMFT architecture is build based on a vision transformer (VIT) [36] because of its benefits in global context modeling, scalability, parameter efficiency, transferability, interpretability, and surpassing performance compared to traditional CNNs in computer vision tasks [37]. Unlike the linear patch embedding layer commonly used in conventional VIT, NIMFT introduces a 1D-CNN patch embedding strategy with non-linear activation to extract local neighborhood information throughout the sEMG and ACC analysis window and project the window into sequential patches, emphasizing their strong contextual correlation. The resulting patchified inner modal features, augmented with a class token (CLS) and position embedding are processed through multi-level NIMFT encoders. These encoders utilize cross-attention mechanisms to extract specific intramodal information and interactive intermodal knowledge, enabling comprehensive fusion thoroughout the entire encoder module. The overall structure of NIMFT is illustrated in figure 1.

2.1. 1D-CNN embedding layer

In the traditional VIT architecture, input signals are divided into non-overlapping patches and mapped to the input dimension through linear projection. However, modeling the structural information within local patches solely through linear projection can be less effective [38], as it overlooks local relationships and internal structural information within the patches [39]. In the field of video representation learning, one approach is to aggregate pixel-level features for each frame by adding one-dimensional convolution (1D-CNN) to capture temporal clues between the same spatial positions [40]. Attention mechanisms provide the model with the ability to focus on long-range information when dealing with sequential data, and combining 1D-CNN, which focuses on neighborhood information, with attention mechanisms that attend to long-term temporal information can effectively help the model understand different regions within the signal. In some cases, such as in a Kaggle competition [41], it was suggested that when strong inter-frame correlation exists in modeling time-series data, 1D-CNN may be more effective than Transformers. Inspired by these insights, we employ a 1D-CNN with non-linear activation for patch embedding. This approach employs neighborhood convolutions to extract localized interactive information between channels and local contextual information within the analysis

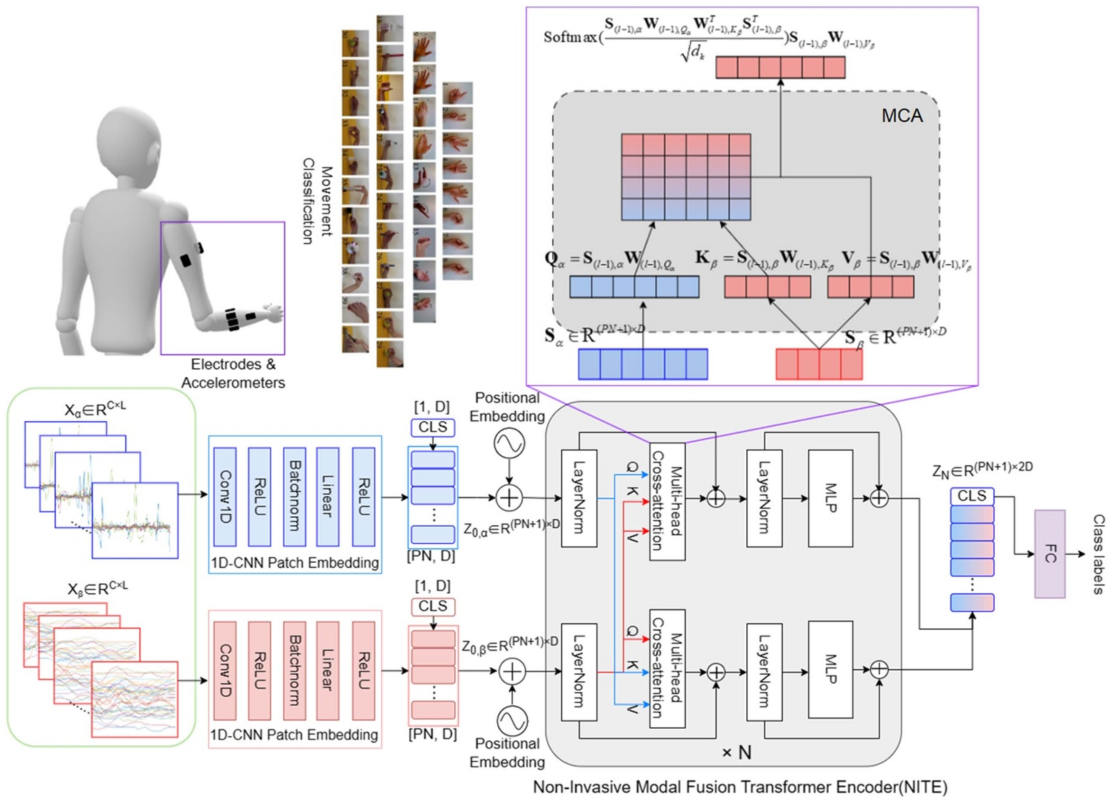

Figure 1. The overall architecture of the NIMFT model. The architecture is based on a transformer encoder framework and consists of the following key modules: patch embedding, which segments input signals into patches using a 1D-CNN and subsequently maps these blocks into unified dimensional tensor representations; position embedding, which is employed to incorporate positional information of each block, including the addition of a class token; modal cross-attention (MCA), which facilitates interaction between dual-modal inputs and seamlessly combines the two input signals along the time dimension.

window. Subsequently, the modal fusion attention module effectively incorporates both short and long-range inter-patch correlations. We define two modalities α (sEMG) and β (ACC). The formula for patch embedding of a specific modality is as follows:

$$\mathbf{Y}_i = \mathrm{ReLU}_2(\mathrm{Linear}_2(\mathrm{ReLU}_1(\mathrm{Linear}_1(\mathrm{Conv1d}(\mathbf{X}_i))))),\quad i = \alpha \text{ or } \beta \tag{1}$$

where $\mathbf{X}_i \in R^{C\times L}$ denotes the resolution of the original signal, C represents the number of channels, and L represents the window length. ReLU_1 and ReLU_2 denote the non-linear activation function ReLU. In contrast to the original VIT, which employs a 2D CNN with a large kernel size and stride equal to the patch size, our approach utilizes a modified configuration with a small kernel size of 1/2 patch size and a stride of 1/4 patch size. After applying 1D-CNN patch embedding, the input signal is converted into input patches denoted as $\mathbf{Y}_i \in R^{\mathrm{PN}\times \mathrm{D}}$, PN represents the resulting number of patches. The implementation of two fully-connected projections with non-linear activation guarantees that the patch size (PS) = L/PN after 1D-CNN, and the patch dimension is converted to D, remaining constant throughout the subsequent transformer encoder. Similar to the BERT framework [42], we append a trainable class token CLS $\mathbf{Z}_0^0 = \mathbf{Y}_{\mathrm{class}} \in R^{1\times \mathrm{D}}$ at the beginning of the embedded patch sequence to capture the meaning of the entire segmented input. Finally, we introduce standard learnable 1D position embedding denoted as $\mathbf{E}_{\mathrm{pos}} \in R^{(\mathrm{PN}+1)\times \mathrm{D}}$ to convey positional information to the Transformer. The final form of embedded patches for each modality is as follows:

$$\mathbf{Z}_{0,i} = \left[\mathbf{Y}_{\mathrm{class}}; \mathbf{Y}_i^1; \mathbf{Y}_i^2; \ldots\ldots; \mathbf{Y}_i^{\mathrm{PN}}\right] + \mathbf{E}_{\mathrm{pos}},\ i = \alpha \text{ or } \beta. \tag{2}$$

2.2. Non-invasive modal fusion transformer encoder

The non-invasive modal fusion transformer encoder takes the $\mathbf{Z}_{0,i}$ as inputs. The NITE consists of N layers. Each layer contains two modules, namely the MCA module and the multi-layer perceptron (MLP) module. Taking the sEMG branch (the upper part of NITE grey box in figure 1) as an example, the encoding process of NITE are as follows:

$$\mathbf{Z}'_{l,\alpha} = \mathrm{MCA}(\mathrm{LayerNorm}(\mathbf{Z}_{(l-1),\alpha})) + \mathbf{Z}_{(l-1),\alpha},\ l = 1\ldots N \tag{3}$$

$$\mathbf{Z}_{l,\alpha} = \mathrm{MLP}(\mathrm{LayerNorm}(\mathbf{Z}'_{l,\alpha})) + \mathbf{Z}'_{l,\alpha},\ l = 1\ldots N. \tag{4}$$

A layer-normalization is used before MCA and MLP modules, and the residual connections are applied to address degradation problem. Specifically, for the sEMG branch, MCA module calculates the query based on the features of the sEMG, meanwhile it incorporates the representation of the ACC branch, which projects the input $\mathbf{Z}_{(l-1),\beta}$ to obtain the key and value.

Let $\mathbf{S}_{(l-1),\alpha} = \mathrm{LayerNorm}(\mathbf{Z}_{(l-1),\alpha}), \mathbf{S}_{(l-1),\beta} = \mathrm{LayerNorm}(\mathbf{Z}_{(l-1),\beta})$, the formulas of MCA are as follows:

$$\mathbf{S}_{l,\alpha} = \mathrm{Softmax}\left(\frac{\mathbf{S}_{(l-1),\alpha}\mathbf{W}_{(l-1),Q_\alpha}\mathbf{W}^{\mathrm{T}}_{(l-1),K_\beta}\mathbf{S}^{\mathrm{T}}_{(l-1),\beta}}{\sqrt{d_k}}\right) \times \mathbf{S}_{(l-1),\beta}\mathbf{W}_{(l-1),V_\beta} \tag{5}$$

where $d_k = D/head$ and the cross-attention matrix A can be calculated as:

$$\mathbf{A} = \mathrm{Softmax}\left(\frac{\mathbf{S}_{(l-1),\alpha}\mathbf{W}_{(l-1),Q_\alpha}\mathbf{W}^{\mathrm{T}}_{(l-1),K_\beta}\mathbf{S}^{\mathrm{T}}_{(l-1),\beta}}{\sqrt{d_k}}\right). \tag{6}$$

The Softmax activation promotes competition in the attention matrix [43], thus highlighting more important attributes and timestamps of each modality. As a result, it provides the importance score of each key relative to each query, that is, the importance of each representation of modality α with respect to modality β. Consequently, the features that exhibit agreement between the two modalities exert the greatest influence on the final prediction, thereby guiding the model to learn features that demonstrate a substantial level of agreement across modalities. Similarly, the MCA module of the ACC branch can be expressed as:

$$\mathbf{S}_{l,\beta} = \mathrm{softmax}\left(\frac{\mathbf{S}_{(l-1),\beta}\mathbf{W}_{(l-1),Q_\beta}\mathbf{W}^{\mathrm{T}}_{(l-1),K_\alpha}\mathbf{S}^{\mathrm{T}}_{(l-1),\alpha}}{\sqrt{d_k}}\right) \times \mathbf{S}_{(l-1),\alpha}\mathbf{W}_{(l-1),V_\alpha} \tag{7}$$

Subsequently, the MLP module of NITE consists of two linear layers in which the first layer is followed by Gaussian error linear unit activation function. After thorough feature fusion in both streams, the outputs of these two streams are concatenated to form a vector $\mathbf{Z}_N \in \mathrm{R}^{(\mathrm{PN}+1)\times 2\mathrm{D}}$. Subsequently, the output corresponding to the class token $\mathbf{Z}_N^0$ is activated using the Softmax function. Finally, in the FC layer, the predicted class labels for each path are calculated based on their respective outputs as follows.

$$\mathbf{Y}_{\mathrm{label}} = \mathrm{Linear}\left(\mathbf{Z}_N^0\right). \tag{8}$$

3. Experiment & results

3.1. Database

The proposed architecture was assessed using a publicly available dataset Ninapro DB2 [44]. The dataset contains synchronously collected 12-channel sEMG (Delsys Trigno wireless acquisition system) and 3-axis ACC signals of 40 healthy subjects when performing 49 distinct hand gestures. The dataset is categorized into three sets: Exercise B (9 basic wrist movements and 8 isometric/isotonic hand configurations), Exercise C (23 grasp and functional movements), and Exercise D (9 finger force patterns). Each subject performed six repetitions of each gesture, with each repetition lasting 5 s, followed by a 3-second rest period. Both sEMG and ACC are presented at a sampling rate of 2000 Hz when the dataset was published.

3.2. Experimental setting

(1) Processing Enviroment: The experiments in this study were performed on an NVIDIA GeForce RTX 3060 GPU platform with Python 3.6, and the proposed model is implemented with Pytorch.

(2) Data preprocessing: Raw sEMG signals were first high-pass filtered (1st Butterworth filter, 20 Hz), and then normalized using a μ-law normalization technique. This normalization approach aimed to amplify the small amplitude sensor channels while preserving the scale of sensors with large amplitude [6]. For a given input signal $\mathbf{X}_t$, normalized signal is as follows:

$$\mathrm{F}(\mathbf{X}_t) = \mathrm{sign}(\mathbf{X}_t)\frac{\ln(1+\mu|\mathbf{X}_t|)}{\ln(1+\mu)} \tag{9}$$

where the signal range is defined by the parameter $\mu = 256$. After normalization, the sEMG signal is segmented using a sliding window with an overlap time of 50 ms. The selection of the window size is discussed in section 3.3. Each output of the segmentation phase is represented by $\mathbf{X} \in R^{C\times L}$,where L represents samples within each window, and C is the number of channel (C = 12 in this study). For 3-axis ACC data, a 1st-order Butterworth high-pass filter with cutoff freqency of 20 Hz same as used in EMG preprocessing was used to remove the low frequency offset induced by slow body movement interferences, while preserving the crucial ACC band during muscle contractions (10–30 Hz) [45]. Then, a μ-law normalization method and a sliding window procedure same as sEMG processing were used.

(3) Evaluation Metric: In sEMG-based single-modal gesture recognition, the typical evaluation method involves training a subject-specific model using a training set that encompasses about two-thirds of each subject's gesture trials. The evaluation is then performed on the remaining one-third of the trials, which constitutes the test set. This approach is known as the intra-subject method [6]. This study aimed at developing a universal model and assessing its

Table 1. The impact of model parameters on accuracy.

	Win size	(PS,PN)	Encoder Layer	Model dimension	Head	Parameter	Accuracy
Encoder layer; Dimension; Head	200 ms	[20, 20]	2	256	8	1696 913	91.96% ± 3.57%
	200 ms	[20, 20]	3	512	8	9772 817	93.73% ± 2.86%
	200 ms	[20, 20]	2	512	4	6621 457	93.71% ± 2.42%
	200 ms	[20, 20]	2	512	16	6621 457	93.23% ± 3.08%
	200 ms	[20, 20]	**2**	**512**	**8**	6621 457	93.91% ± 2.37%
(PS,PN)	200 ms	[25, 16]	2	512	8	6650 129	93.69% ± 2.42%
	200 ms	**[20, 20]**	2	512	8	6621 457	93.91% ± 2.37%
	200 ms	[10, 40]	2	512	8	6570 257	93.95% ± 2.31%
	200 ms	[4, 100]	2	512	8	6564 113	94.27% ± 2.21%
Win size	50 ms	[20, 5]	2	512	8	6613 777	92.5% ± 2.63%
	100 ms	[20, 10]	2	512	8	6616 337	93.22% ± 2.53%
	200 ms	[20, 20]	2	512	8	6621 457	93.91% ± 2.37%
	400 ms	[20, 40]	2	512	8	6631 697	95.07% ± 2.23%
	1000 ms	[20, 100]	2	512	8	6662 417	95.56% ± 2.18%

generalization ability across different subjects. Therefore, we used a training set with data from all 40 subjects for model training and adaptation, thereby encompassing multiple subjects to facilitate the creation of a universal model. Subsequently, individual tests were conducted on the remaining unseen test sets for each subject. This evaluation method is referred to as inter-subject evaluation. Specifically, in accordance with Ninapro DB2 recommendations, we combined four repetitions (1st, 3rd, 4th, and 6th) from all 40 subjects for the training set and reserved the remaining two repetitions (2nd and 5th) for the test set for individual test. We trained each model for 100 epochs using a learning rate decay strategy. The initial learning rate was set to 0.01, with a decay rate of 0.001.

(4) Statistical analysis: The means of two independent groups were compared using independent t-tests, while differences among multiple groups were assessed using one-way ANOVA with multiple comparisons. Significance level was set as 0.05. Descriptive statistics are presented using the mean ± standard deviation (SD) format. The coefficient of variation (CV) is used to compare the degree of variation in different data sets. All analyses were carried out using SPSS 22.0.

3.3. Hyperparameters selection

We first explores the impact of the number of encoder layers, the model dimension of NITE, and the number of attention heads on the model performance, as shown in table 1. When the number of Encoder layers in the model increased from 2 to 3, the model exhibited overfitting. Therefore, the number of encoder layers was set to 2 in the next analyses. Increasing the dimension from 256 to 512 significantly increased the model parameters and also led to a significant improvement in accuracy, whereas variations in the number of heads did not affect the model parameters, and a setting of 8 heads resulted in improved accuracy. Next, we investigated the impact of patch sizes and patch numbers on the model performance during the patch embedding process. As shown in table 1, the accuracy improves as the number of patch increases. Although a patch number of 100 slightly increases accuracy, it significantly increases computation time (1m43s/epoch when PN = 16 to 3m25s/epoch when PN = 100). Considering both accuracy and time, a patch number of 20 is selected for the following analyses. Regarding analysis window segmentation, prior studies have demonstrated that larger window sizes result in improved HGR performance [6, 46], albeit with higher computational demands. Recent studies have found that smaller windows can capture transient EMG features [47], and accuracy drops can be compensated by increasing the number of channels [48]. Therefore, in this study, we evaluated the model's performance using various analysis window sizes (50 ms, 100 ms, 200 ms, 400 ms, and 1000 ms). To ensure real-time HGR with smooth features and increased training observations, a sliding window with steps of 50 ms (notably, 30 ms for the 50 ms window) was used. As presented in table 1, although HGR accuracy increases with longer window size, a trade-off between HGR accuracy and recognition delay, needs to be considered. Therefore, a 200 ms analysis window with a 50 ms overlapping, similar to previous works was chosen. In summary, the following configuration was chosen for the subsequent experiments: a 2-layer NITE with a dimension of 512, [PS, PN] of [20, 20] and a window size of 200 ms.

Additionally, to evaluate NIMFT's robustness, we divided data from 6 repetitions into training and testing sets with ratios of 3:3, 4:2, and 5:1. The average HGR accuracy of NIMFT under 3:3, 4:2, and 5:1 data splitting conditions was 90.23%, 94.08%, and 94.19%, respectively, as illustrated in figure S1 in the supplementary information. Analysis of the learning

Table 2. The impact of embedding method on accuracy.

Datasets	Embedding method	Accuracy
DB2-ExerciseB	Linear	91.90% ± 2.84%
	1D-CNN	**93.91% ± 2.37%**
DB2-ExerciseC	Linear	89.06% ± 4.88%
	1D-CNN	**91.02% ± 3.84%**
DB2-ExerciseD	Linear	94.92% ± 3.31%
	1D-CNN	**95.56% ± 3.02%**

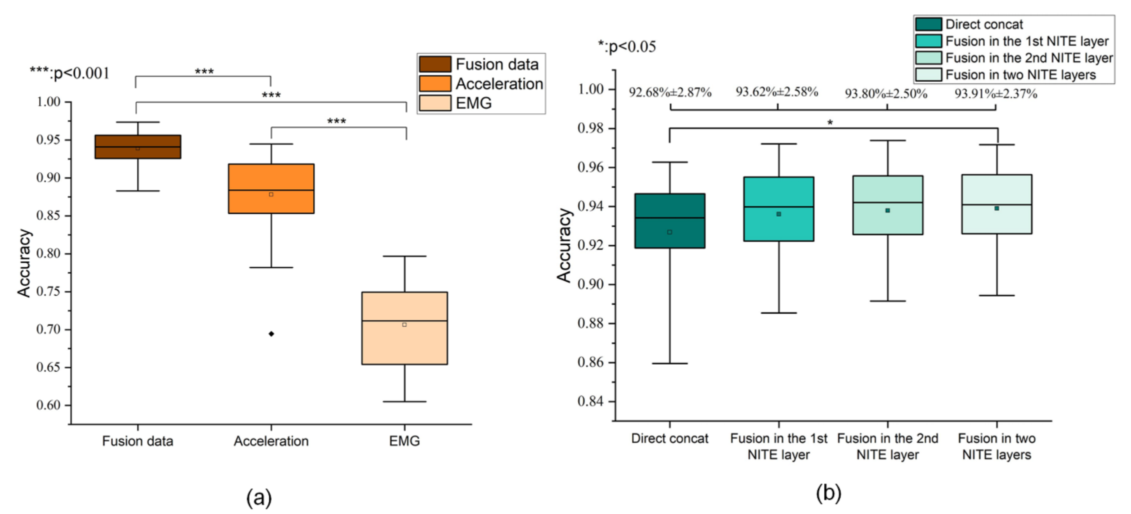

Figure 2. Impact of fusion methods on test accuracy. (a) The average test accuracy on 40 subjects using unimodal and multimodal data. (b) The average test accuracy for four different fusion methods, where there exists a significant difference between direct concat and fusion in two NITE layers ($p < 0.05$).

curve (figure S2) indicated that the pronounced accuracy decline under the 3:3 split may be attributed to overfitting. Although high accuracies (over 94%) were achieved with 4:2 and 5:1 splits, we observed differences across strategies. These may stem from signal amplitude variability among subjects during different movement repetitions, as highlighted in Ninapro DB2's original technical validation [44]. As a result, we adopted the data splitting methodology recommended in the 'Evaluation Metric' section for subsequent evaluations. This method ensures a fair comparison with most recent DL models developed using Ninapro DB2 data [6, 10, 23, 30].

3.4. Ablation study

3.4.1. 1D-CNN patch embedding vs. linear patch embedding

In the presented architecture, a 1D-CNN non-linear embedding was implemented to enhance localized contextual information extraction, and its performance was evaluated against the linear embedding of the original VIT across three action sets within the DB2 dataset. As shown in table 2, the 1D-CNN approach consistently exhibited superior recognition accuracy and a smaller standard deviation than linear embedding across all three action sets. The 1D-CNN-based embedding demonstrated enhanced generalization capabilities across diverse subjects.

3.4.2. Unimodal HGR vs. multimodal HGR

To evaluate the efficacy of fused data in HGR, we conducted comparative experiments using unimodal and multimodal data on Exercise B of the DB2 dataset. Figure 2(a) demonstrates that the fused data approach yielded a significant improvement, achieving an average accuracy of 93.91%, compared to using only sEMG signals (70.64%, $P < 0.001$) or ACC signals (87.70%, $P < 0.001$). Additionally, notable variations in the test outcomes were observed when employing unimodal signals independently across subjects. The calculation of the CV for the test accuracy of 40 subjects revealed values of 7.87% for sEMG, 5.69% for ACC, and 2.52% for fused data. This indicates that fused data not only improves overall accuracy but also reduces performance variability among subjects, resulting in more stable and consistent classification outcomes.

3.4.3. MCA modal fusion vs. invasive modal fusion

To assess the potential improvements offered by NITE with non-invasive MCA fusion module, we compared four fusion approaches: 1. Direct concatenation: modalities are processed through two separate transformer encoders before being directly concatenated; 2. Fusion in the 1st NITE Layer: modalities first go through a modal fusion NITE layer, then proceed through separate transformer encoders; 3. Fusion in the 2nd NITE Layer: Modalities are initially processed by separate transformer encoders, with fusion occurring in the NITE layer afterward. 4. Fusion in Two NITE Layers: Modalities pass through two successive NITE layers. Figure 2(b) showed that the three fusion approaches using NITE layers achieved higher accuracy and lower standard deviation compared to the direct concatenation method. Moreover, employing two layers of NITE achieved the highest recognition accuracy of ($93.91 \pm 2.37\%$). One-way ANOVA post-hoc analysis indicated a significant improvement over direct concatenation ($p = 0.035$).

3.5. Transfer learning

The inherent variability in sEMG and ACC signals across different subjects limits the generalization ability of subject-specific HGR models, affecting their applicability in intersubject scenarios. Even in user-specific contexts, the acquisition of a substantial volume of training data is imperative to ensure the effectiveness of DL-based HGR models, which can impose a significant training burden on users. In this study, our goal is to employ transfer learning with a generic NIMFT model for rapid adaptation to new subjects. We begin by transferring weights from a pre-trained model trained on 39 subjects then fine-tuning it with data from the new subject, which we refer to as leave-one-subject-out (LOSO) fine-tuning [32]. We assess knowledge transfer by comparing subject-specific and TL models, focusing on whether TL fine-tuning surpasses subject-specific training in accuracy and improves HGR with less data collection and training time. All TL experiments are conducted using Exercise B from the DB2 dataset.

3.5.1. Subject-specific vs fine-tuning

In this context, the fine-tuning model was created through the LOSO fine-tuning method, while the subject-specific model denotes a model trained exclusively for a particular individual. Both models adopt a training-to-testing data ratio of 2:1. As shown in figure 3, transferring weights from the pre-trained model has a significant impact on the training dynamics. The fine-tuning model starts at a higher point in validation accuracy and reaches saturation accuracy after fewer epochs. In contrast, the training process of the subject-specific model is relatively slow and requires more epochs to reach a similar level of accuracy. Moreover, the fine-tuning model surpasses the subject-specific model in terms of final validation accuracy, exhibiting reduced inter-subject variability. Transferred knowledge and features may offer an improved starting point for a new subject model, making it easier to find the global optimum during training and achieve higher performance levels.

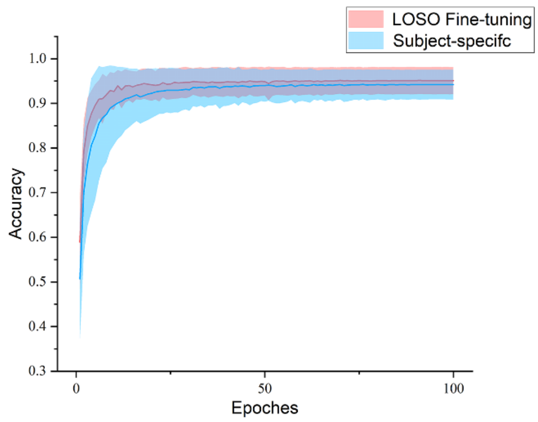

Figure 3. Validation accuracy of subject-specific and fine-tuning models. Subject-specific models begin with lower accuracy and take longer time to reach saturation. Fine-tuning models start higher and typically achieve better final accuracy.

3.5.2. The amounts of samples affect on retraining

In practical applications of deep learning models, achieving shorter training times for individual subjects is a desirable objective, which can be realized by expediting convergence and reducing the volume of training data. As seen in figure 3, fine-tuning converges earlier compared to subject-specific training. To assess the capacity of fine-tuning of NIMFT model in improving performance with shorter training duration and reduced data volume, we employed an early stopping evaluation approach. In this method, fine-tuning was terminated when a stable state was reached, defined as a loss reduction of less than 0.001 over ten consecutive epochs. We then compared accuracy at the same epoch between fine-tuning and subject-specific training. Additionally, we systematically varied the number of motion repetitions in the dataset for retraining to determine the optimal repetition level for achieving acceptable HGR accuracy. As shown in figure 4, the accuracy of the pretrained LOSO model without fine-tuning stands at only $21.46\% \pm 5.78\%$. When fine-tuned with varying repetitions, the pretrained model outperforms the subject-specific model. In the 2–5 repetitions of motion range, at early-stop epochs of 50(1 min9s, 1.37 s/epoch), 53(1 min54s, 2.15 s/epoch), 55(2 min37s, 2.86 s/epoch), and 57(3 min21s, 3.53 s/epoch), the median success rates for fine-tuning and subject-specific recognition were as follows: 87.96% vs. 86.28%, 87.53% vs. 86.42%, 95.83% vs. 93.57%, and 96.13% vs. 94.65%. T-test further confirmed that when 4 or 5 repetitions were employed for retraining, the accuracy of the fine-tuned model was significantly higher than that of the subject-specific model with $p = 0.033$ and $p = 0.002$, respectively. This demonstrates that fine-tuning effectively utilizes the pre-trained model's knowledge, leading to superior performance even with less training data.

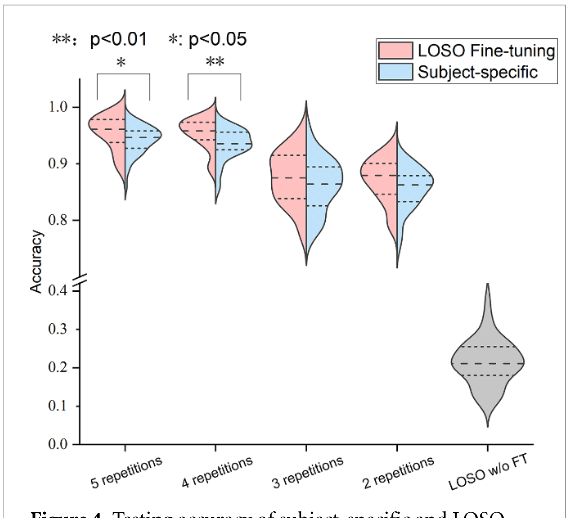

Figure 4. Testing accuracy of subject-specific and LOSO fine-tuning models with varying numbers of samples for retraining. Overall, fine-tuning outperforms subject-specific, and the improvement with fine-tuning decreases when the number of repetitions used for retraining is reduced. When 4, 5 repetitions used for retraining, there was a significant difference between the two methods.

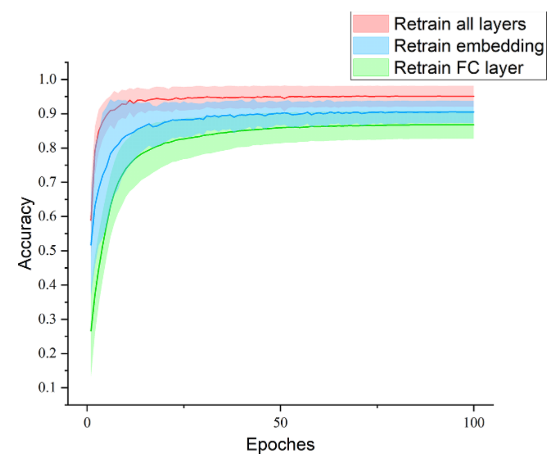

Figure 5. Impact of retraining different layers on validation accuracy. Three methods were employed in this study: (1) retraining all layers; (2) retraining only the FC layer; (3) retraining only the embedding layer.

3.5.3. Retrain different layers

When adapting the model to new subjects, selectively freezing a portion of the network's layers and fine-tuning specific layers can enhance training efficiency [32]. In our study, we explored three retraining strategies: 1. Retraining all layers; 2. Retraining the embedding layers before NIMFT module; 3. Retraining the final FC layer. As shown in figure 5, retraining all layers produced the highest recognition accuracy for our task and dataset, achieving an accuracy of 95.10% ± 3.05% after 100 epochs. Retraining the embedding layers resulted in an accuracy of 90.50% ± 3.27%, while retraining the final FC layer achieved an accuracy of 86.77% ± 4.07%. This observation further emphasizes the model's dependence on valuable subject-specific cross-modality features from the NITE module, underscoring the need for their retraining during the fine-tuning process.

3.6. Models compare

Recent studies have utilized DL models to fuse sEMG and ACC signals from sparse multi-channel sensors for improved HGR. The majority of these approaches combine traditional feature extraction with DNNs, with subject-specific assessments [10, 30, 49, 50]. Notably, state-of-the-art HGR methods using end-to-end sEMG and ACC DL models, such as NIDA-TCN [31] and two-stream LSTM-Res [35], lack fair comparative evaluations on public databases like Ninapro. Hence, we replicated these models for comparative analysis with our proposed NIMFT on the Ninapro DB2. NIDA-TCN [31], denoting the non-invasive bimodal attention temporal convolutional network, was initially employed for the recognition of daily ambulatory activities. As illustrated in figure 6(a), this approach entails seperate processing of sEMG and ACC signals via four TCN layers. Subsequently, they are non-invasively fused within the temporal attention module (TAM) and channel attention module (CAM), ultimately culminating in a final prediction achieved through a FCN. The Two-stream RNN-Res model was employed to identify 20 hand and upper extremity activities using MYO armband [35]. As depicted in figure 6(b), this model integrates the extraction of long-term dependency features through LSTM with the extraction of local features via CNN. It also incorporates a residual network to mitigate network degradation, and finally, it fuses the two modalities through direct concatenation. Significantly, the original NIDA-TCN and two-stream LSTM-Res models employed analysis windows of 200 ms and 1000 ms, respectively. To ensure equitable comparisons, we selected a 400 ms window in this study. We also listed comparisons between NIMFT and non-end-to-end HGR methods reported in the existing literature [10, 30, 49, 50] in table 3. In conclusion, our overall recognition accuracy achieved 91.92%, establishing NIMFT as the top-performing end-to-end model in terms of recognition accuracy when employing inter-subject training and evaluation. While three models did report higher accuracy compared to our model, it is worth noting that they conducted training and evaluation on a subject-specific basis. Additionally, we deliberately omitted the 'rest' class from our experiment, as it constitutes a significant portion of the dataset, resulting in serious class imbalance.

In particular, one-way ANOVA revealed significant improvements in classification accuracy for our model over the Two-stream LSTM-Res and NIDA-TCN models ($p = 7.09 \times 10^{-7}$ and 3.11×10^{-4}). Figure 7(b) demonstrates the superior performance of our model compared to the other two in terms

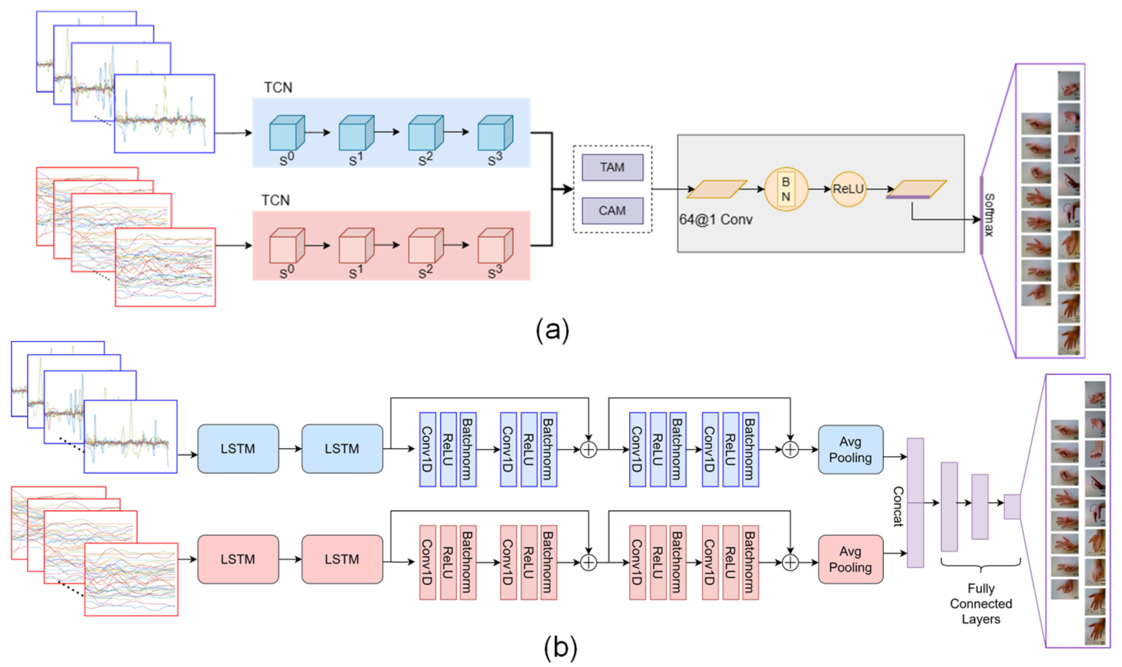

Figure 6. Two end-to-end model used for hand gesture recognition. (a) NIDA-TCN model architecture. (b) Two-stream LSTM-Res model architecture.

Table 3. Comparison with state-of-the-art models on Ninapro DB2.

Method	Datasets	Modalities	Num of gestures	Win- size	End-to-end?	Inter-subject?	Accuracy
KRLS [49]	DB2	sEMG-ACC	40	400 ms	No	No	82.49%
Improved KRLS [50]	DB2	sEMG-ACC	40	400 ms	No	No	92.10%
MV-CNN [10]	DB2	sEMG-ACC	50	200 ms	No	No	94.40%
HyFusion-summation [30]	DB2	sEMG-ACC	50	200 ms	No	No	**94.73%**
Two-stream LSTM-Res [35]	DB2	sEMG-ACC	49	400 ms	Yes	Yes	87.94%
NIDA-TCN [31]	DB2	sEMG-ACC	49	400 ms	Yes	Yes	85.77%
NIMFT	DB2	sEMG-ACC	49	400 ms	Yes	Yes	**91.92%**

of accuracy, precision, recall, and F1-score, while maintaining comparable model complexity (Number of parameters = 6631 697, 6990 530, and 6372 561, respectively). It is essential to emphasize that the Two-stream LSTM-Res and NIDA-TCN models were faithfully implemented using their original architectures and optimal hyperparameters as presented in their respective papers, without any customized tuning for the Ninapro DB2 dataset.

4. Discussion

In pursuit of improving HGR performance and fostering cross-subject transferability, our study introduces an end-to-end transferable modal fusion HGR model based on raw sEMG and ACC Signals. This model incorporates a 1D-CNN based embedding module along with a non-invasive NITE, aimed at addressing the inherent challenges in traditional modal fusion techniques, such as complex modal feature engineering [12, 14] and vulnerability to contamination in invasive modal fusions [31].

4.1. Patch embedding with 1D-CNN

The original ViT models substitute the inherent inductive bias favoring local processing in convolutions with global processing facilitated by multi-head self-attention, resulting in enhanced performance in vision tasks. Recently, Montazerin introduced the ViT-HGR framework, leveraging the ViT architecture to efficiently classify 65 hand gestures using HD-sEMG data with a significant accuracy of 84.6% [51], demonstrating its effectiveness in processing sequential signals like sEMG [51]. However, it is important to note that the patchification using linear projection in the original ViT may lead to the loss of neighborhood dependencies within patches, particularly in closely related sequential signals. Inspired by Xiao *et al*'s [52] recommendation to integrate early convolutional layers with 3×3 kernels and a small stride into the ViT model to balance localized inductive biases and transformer block capabilities while improving optimization stability, our study employs a patch embedding module based on temporal 1D-CNN with non-linear projection. This approach

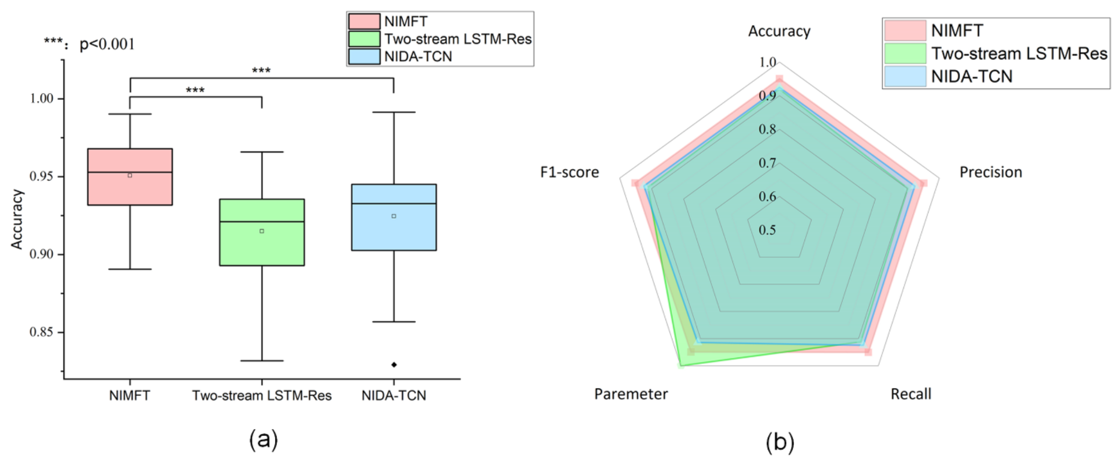

Figure 7. The performance of the proposed model (NIMFT), Two-stream LSTM-Res and NIDA-TCN models. (a) Average accuracy of three models on DB2 Exercise B with a 400 ms window. (b) Prediction accuracy, recall, precision, F1-score, and model parameters of three models, respectively.

allows each sequence element to capture its nearby elements, enhancing short-range feature representation for subsequent long-range transformer architecture. In the ablation study, the incorporation of 1D-CNN embedding had a significantly positive impact on model performance and generalization capabilities. Across all three action sets within Ninapro DB2, our 1D-CNN embedding consistently outperformed linear patch embedding. Notably, when handling data from different subjects, our model exhibited superior performance. For instance, during the evaluation of Exercise C in DB2, the lowest subject-specific test accuracy was 69.88% with linear patch embedding, but it improved to 79.56% when 1D-CNN patch embedding was utilized. This observation suggests that the feature sequences of sEMG and ACC learned by 1D-CNN more effectively capture inter-subject differences in temporal characteristics, thereby enhancing the model's adaptability to unseen data.

4.2. Effects of various fusion methods on multi-modal HGR

To seamlessly integrate homogeneous and heterogeneous information within multimodal data in a non-invasive manner, we introduced the MCA mechanism, merging data across different modalities using a query-key-value system. By computing query-key similarity, attention weights between modalities were adjusted through every layer of the NITE module to better accommodate their distinct features, further promoting cross-modal information exchange. As depicted in figure 2(b), the use of two NITE modules resulted in an average recognition accuracy increase of 1.23% compared to direct concatenation. We also noted that employing a single NITE, specifically in the first layer, led to a performance improvement of 0.98%, while in the second layer, it yielded a 1.12% enhancement in comparison to direct concatenation. To provide a deeper insight into the underlying enhancement mechanisms enabled by the MCA-based NITE, we conducted attention score visualizations and examined their distribution. As illustrated in figure 8(a), taking the MCA in the second NITE module as an example, we observed that the rightmost cross-attention map (post-softmax) exhibits a greater concentration of high attention scores in specific areas, contrasting with the attention map of other single-modality configurations involving two distinct multi-head self-attention mechanisms. The attention score distribution in figure 8(b) demonstrates a broader range with both higher and lower values compared to single-modal attention. The upper quantile has shifted from 0.0486 (EMG only) and 0.0484 (ACC only) to 0.0503 (MCA). These findings illustrate that the MCA has acquired the capability to focus on meaningful signals spanning both modalities, leading to a more effective capture of cross-modal events over long distances, ultimately enhancing the performance of HGR.

4.3. Model transferability validation

This study rigorously evaluates the effectiveness of NIMFT-based transfer learning, echoing recent work by Lehmler *et al* [32]. We delve into the performance comparisons between subject-specific and fine-tuning models, focusing on training dynamics, training data volume impact, and layer-specific fine-tuning effects. We achieved superior overall accuracy enhancement through transfer learning compared to the previously reported deep transfer learning HGR method using only sEMG [32, 34]. When applying TL to Ninapro DB2, the detailed accuracy comparison with and without TL yielded results of 94.65% to 96.13% in our approach, 49.23% to 67.89% in the study by Lehmler *et al* [32], and 63.73% to 65.29% in the work of Zou and Cheng [34]. Specifically,

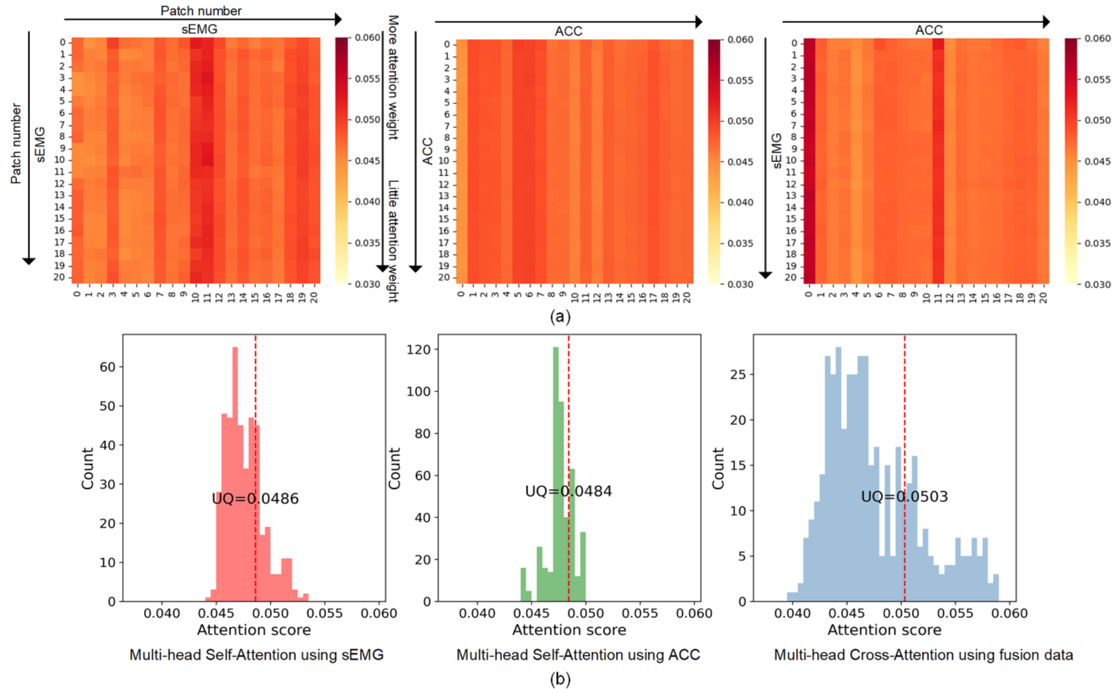

Figure 8. Attention score under different conditions. (a) From left to right: attention scores of sEMG with two multi-head self-attention, attention scores of ACC with distinct multi-head self-attention, attention scores of bimodal fusion with two multi-head cross-attention. Since a CLS token was introduced in the embedding stage, there were 21 patch number in each 200 ms window. Therefore, the attention map size was 21 × 21. (b) Histograms of the corresponding values of the three attention score heatmaps. The upper quantile (UQ) lines were shown in the figure (red dashed lines).

Stephan noted a substantial gap in transfer learning improvement when using an end-to-end model with raw sEMG compared to manual sEMG feature extraction. In contrast, our study demonstrates that end-to-end NIMFT using raw signals with TL achieves notably high accuracy. This phenomenon may be attributed to two key factors. Firstly, the training dynamics, as illustrated in figure 3 with LOSO fine-tuning approach, highlight the rapid increase in our model's accuracy from 58.88% to over 90% within only 5 epochs. Our model demonstrates a noteworthy capacity for acquiring generalized feature representations, even in the presence of substantial inter-subject variations within the Ninapro DB2 dataset. Secondly, prior researches have shown that sEMG data display greater inter-class feature ambiguity when compared to ACC data [31, 53]. The stability of ACC features, in conjunction with multi-layer cross-attention mechanisms capable of extracting intermodal similarity, is likely responsible for mitigating the ambiguity in sEMG features and enhancing the transferability of our model.

In this study, we explored strategies to reduce fine-tuning training time by fine-tuning specific layers and varying the number of training samples. Fine-tuning only the final prediction layer is a commonly adopted approach in deep TL, assuming that preceding layers have already captured general features. However, for sEMG signals, retraining previous layers may also be advantageous, especially in scenarios involving electrode displacement [54] and muscle state variability [55] during movements. Unlike Stephan´s research [32], which did not observe benefits from training all layers compared to training only the first or last layer, our study revealed that retraining all layers yields higher accuracy than retraining just the embedding layer, with the worst results obtained when retraining only the last layer. This suggests that our model has effectively acquired valuable features, particularly in the MCA section, which significantly improved HGR performance. Notably, retraining the embedding layer exerted a more pronounced influence compared to retraining the final FC layer. This can be attributed to the substantial inter-subject variations and dissimilarity between the source and target domains. Retraining the preceding layers facilitates improved adaptation of new subjects to the source domain's feature space.

4.4. Comparison with state-of-the-art models

It is worth noting that major differences between HGR studies, such as different data acquisition protocols, number of movements, type of movements, and evaluation procedure, could restrict a fair comparison, especially in recently reported sEMG-IMU modal fusion studies. In this study, to ensure fair comparison we reconstructed two state-of-art end-to-end sEMG-ACC fusion model with the reported optimal hyperparameters and evaluating them against NIMFT using inter-subject assessments on the

Ninapro DB2 dataset. Through the refinement of the VIT model involving a locally contextual embedding layer and an non-invasive MCA strategy tailored to the characteristics of sequential sEMG and ACC signals, our NIMFT approach outperformed NIDA-TCN and Two-stream LSTM-Res in all four evaluation metrics, while maintaining a comparable model size. Furthermore, the scalability and proficiency in modeling long-range dependencies inherent in the VIT architecture were preserved. However, it is essential to note that we did not conduct exhaustive tuning for these models specifically on the Ninapro DB2 dataset. For instance, in [31], a comprehensive grid search was carried out to assess various input combinations for TAM and CAM, and we chose the input combination that had demonstrated the best performance in that study.

Two studies listed in table 3 merit attention [10, 30]. These models exhibit superior overall accuracy compared to NIMFT across the entire Ninapro DB2 dataset. It is noteworthy that these models employ an extensive feature extraction approach, exemplified by the Hyfusion model, which manually extracts 12 sEMG features and 6 ACC features. In addition, these models were trained and evaluated using an intra-subject paradigm. Notably, the transferable evaluation in our study showed the potential advantagement of training a general model using inter-subject datasets and transfer to an unseen subject can speed up model convergence and achieve a higher accuracy than intra-subject training. Therefore, for practical considerations aimed at reducing the training burden on users, it is advisable to prioritize inter-subject evaluations to assess model generalization.

4.5. Study limitations and future directions

As shown in figure 4, the pre-trained LOSO model achieved an accuracy of 21.46% $\pm$ 5.78% without fine-tuning, slightly surpassing the accuracy below 20% reported by Wei *et al* [10] for a multi-view deep learning model using only EMG data from Ninapro DB2. In addition, 4 or 5 repetitions of retraining data should be used to achieve a genuinely practical HGR system with an average accuracy exceeding 95%. This results in increased training time and places a higher burden on subjects due to the need for more sample repetitions. In future research, incorporating domain adaptation methods into the NIMFT architecture, such as domain adversarial learning [56], may enhance the alignment of features between the source and target domains, thereby improving the generalization capability of earlier feature extraction layers. Additionally, the integration of few-shot learning strategies could be considered, aiming to predict the desired output with minimal training observations, thereby substantially reducing the user burden [6].

Besides, the 'rest' class accounts for more than 30% of the entire Ninapro DB2, potentially leading to class imbalance concerns. A prior study by Fatayer *et al* [57] identified pronounced label noise primarily associated with the erroneous labeling of the 'rest-movement-rest' sequence. Consequently, including the 'rest' class may introduce model bias towards the majority class, potentially leading to misinterpretations of model performance. To address this, similar to several HGR studies [23, 58], we intentionally omit the 'rest' class, incorporating only 49 hand gestures for horizontal comparative analysis alongside NIDA-TCN and Two-stream LSTM-Res. In future research, the refinement of the 'rest'; label using movement detection strategies, such as Akram's ALR CNN, should be considered in the model evaluation process to address class imbalance.

5. Conclusion

In this study, we present NIMFT, an innovative end-to-end multi-modal HGR model. NIMFT improves recognition accuracy and model stability by employing 1D-CNN-based patch embedding and non-invasively integrating information from both sEMG and ACC data through an MCA-based NITE module. Additionally, NIMFT supports seamless adaptation to new subjects via transfer learning. Our results demonstrate NIMFT's superiority over the most recent end-to-end bimodal fusion models, achieving an impressive inter-subject classification accuracy of 91.92% for 49 gestures in Ninapro DB2. This model holds substantial promise for the advancement of personalized HMI devices and rehabilitation strategies in future research.

Data availability statement

A publicly available dataset was used in this study. The data that support the findings of this study are openly available at the following URL/DOI: https://ninapro.hevs.ch/instructions/DB2.html.

Acknowledgments

This research was funded by the Leading-edge Technology and Basic Research Program of Jiangsu (Grant No. BK20192004D), the Key Research and Development Program of Jiangsu (Grant No. BE2022160), the Natural Science Foundation of Jiangsu Province (BK20230301, BK20230302) and the National Key Research and Development Program of China (Grant No. 2022YFC2405600).

ORCID iD

Yuxuan Zhou https://orcid.org/0000-0002-0432-1720

References

[1] Guo L, Lu Z and Yao L 2021 Human-machine interaction sensing technology based on hand gesture recognition: a review *IEEE Trans. Hum.* **51** 300–9

[2] Wu X, Yuan Y, Zhang X, Wang C, Xu T and Tao D 2021 Gait phase classification for a lower limb exoskeleton system based on a graph convolutional network model *IEEE Trans. Ind. Electron.* **69** 4999–5008

[3] Lee S, Kim M-O, Kang T, Park J and Choi Y 2018 Knit band sensor for myoelectric control of surface EMG-based prosthetic hand *IEEE Sens. J.* **18** 8578–86

[4] Yu Y, Chen X, Cao S, Zhang X and Chen X 2019 Exploration of Chinese sign language recognition using wearable sensors based on deep belief net *IEEE J. Biomed. Health Inf.* **24** 1310–20

[5] Zhou Y-X, Xia Y, Huang J, Wang H-P, Bao X-L, Bi X-Y, Chen X-B, Gao Y-J, Lü X-Y and Wang Z-G 2017 Electromyographic bridge for promoting the recovery of hand movements in subacute stroke patients: a randomized controlled trial *J. Rehabil. Med.* **49** 629–36

[6] Rahimian E, Zabihi S, Asif A, Farina D, Atashzar S F and Mohammadi A 2021 FS-HGR: few-shot learning for hand gesture recognition via electromyography *IEEE Trans. Neural Syst. Rehabil. Eng.* **29** 1004–15

[7] Geng W, Du Y, Jin W, Wei W, Hu Y and Li J 2016 Gesture recognition by instantaneous surface EMG images *Sci. Rep.* **6** 36571

[8] Clarke A K, Atashzar S F, Vecchio A D, Barsakcioglu D, Muceli S, Bentley P, Urh F, Holobar A and Farina D 2021 Deep learning for robust decomposition of high-density surface EMG signals *IEEE Trans. Biomed. Eng.* **68** 526–34

[9] Naik G R, Al-Timemy A H and Nguyen H T 2016 Transradial amputee gesture classification using an optimal number of sEMG sensors: an approach using ICA clustering *IEEE Trans. Neural Syst. Rehabil. Eng.* **24** 837–46

[10] Wei W, Dai Q, Wong Y, Hu Y, Kankanhalli M and Geng W 2019 Surface-electromyography-based gesture recognition by multi-view deep learning *IEEE Trans. Biomed. Eng.* **66** 2964–73

[11] Ergeneci M, Gokcesu K, Ertan E and Kosmas P 2018 An embedded, eight channel, noise canceling, wireless, wearable sEMG data acquisition system with adaptive muscle contraction detection *IEEE Trans. Biomed. Circuits Syst.* **12** 68–79

[12] Khushaba R N, Al-Timemy A H, Al-Ani A and Al-Jumaily A 2017 A framework of temporal-spatial descriptors-based feature extraction for improved myoelectric pattern recognition *IEEE Trans. Neural Syst. Rehabil. Eng.* **25** 1821–31

[13] Hudgins B, Parker P and Scott R N 1993 A new strategy for multifunction myoelectric control *IEEE Trans. Biomed. Eng.* **40** 82–94

[14] Li W, Shi P and Yu H 2021 Gesture recognition using surface electromyography and deep learning for prostheses hand: state-of-the-art, challenges, and future *Front. Neurosci.* **15** 621885

[15] Bi L, Feleke A G and Guan C 2019 A review on EMG-based motor intention prediction of continuous human upper limb motion for human-robot collaboration *Biomed. Signal Process. Control* **51** 113–27

[16] Yaxu X, Xiaofei J, Dalin Z, Jing L and Zhaojie J 2019 SEMG-based human in-hand motion recognition using nonlinear time series analysis and random Forest *IEEE Access* **7** 176448–57

[17] Pizzolato S, Tagliapietra L, Cognolato M, Reggiani M, Muller H and Atzori M 2017 Comparison of six electromyography acquisition setups on hand movement classification tasks *PLoS One* **12** e0186132

[18] Parvathy P, Subramaniam K, Prasanna Venkatesan G, Karthikaikumar P, Varghese J and Jayasankar T 2021 Development of hand gesture recognition system using machine learning *J. Ambient Intell. Humaniz Comput.* **12** 6793–800

[19] Chen L, Fu J, Wu Y, Li H and Zheng B 2020 Hand gesture recognition using compact CNN via surface electromyography signals *Sensors* **20** 672

[20] Paul M C, Sarkar S, Rahman M M, Reza S M and Kaiser M S 2016 Low cost and portable patient monitoring system for e-health services in bangladesh *2016 Int. Conf. on Computer Communication and Informatics (ICCCI)* pp 1–4

[21] Ameur S, Ben Khalifa A and Bouhlel M S 2020 A novel hybrid bidirectional unidirectional LSTM network for dynamic hand gesture recognition with leap motion *Entertain. Comput.* **35** 100373

[22] Toro-Ossaba A, Jaramillo-Tigreros J, Tejada J C, Peña A, López-González A and Castanho R A 2022 LSTM recurrent neural network for hand gesture recognition using EMG signals *Appl. Sci.* **12** 9700

[23] Shen S, Wang X, Mao F, Sun L and Gu M 2022 Movements classification through sEMG with convolutional vision transformer and stacking ensemble learning *IEEE Sens. J.* **22** 13318–25

[24] Cote-Allard U, Campbell E, Phinyomark A, Laviolette F, Gosselin B and Scheme E 2020 Interpreting deep learning features for myoelectric control: a comparison with handcrafted features *Front. Bioeng. Biotechnol.* **8** 158

[25] Hogrel J-Y, Duchêne J and Marini J-F 1998 Variability of some SEMG parameter estimates with electrode location *J. Electromyogr. Kinesiol.* **8** 305–15

[26] Li Z, Zhao X, Liu G, Zhang B, Zhang D and Han J 2021 Electrode shifts estimation and adaptive correction for improving robustness of sEMG-based recognition *IEEE J. Biomed. Health Inf.* **25** 1101–10

[27] Nordander C, Willner J, Hansson G A, Larsson B, Unge J, Granquist L and Skerfving S 2003 Influence of the subcutaneous fat layer, as measured by ultrasound, skinfold calipers and BMI, on the EMG amplitude *Eur. J. Appl. Physiol.* **89** 514–9

[28] Fan Z and He H 2013 Source selection for real-time user intent recognition toward volitional control of artificial legs *IEEE J. Biomed. Health Inf.* **17** 907–14

[29] Rizzoglio F, Pierella C, De Santis D, Mussa-Ivaldi F and Casadio M 2020 A hybrid body-machine interface integrating signals from muscles and motions *J. Neural Eng.* **17** 046004

[30] Duan S, Wu L, Xue B, Liu A, Qian R and Chen X 2023 A hybrid multimodal fusion framework for sEMG-ACC-based hand gesture recognition *IEEE Sens. J.* **23** 2773–82

[31] Zhou B, Feng N, Wang H, Lu Y, Wei C, Jiang D and Li Z 2022 Non-invasive dual attention TCN for electromyography and motion data fusion in lower limb ambulation prediction *J. Neural Eng.* **19** 046051

[32] Lehmler S J, Saif-Ur-Rehman M, Tobias G and Iossifidis I 2022 Deep transfer learning compared to subject-specific models for sEMG decoders *J. Neural Eng.* **19** 056039

[33] Ketyko I, Kovacs F and Varga K Z 2019 Domain adaptation for sEMG-based gesture recognition with recurrent neural networks *2019 Int. Joint Conf. on Neural Networks (IJCNN)* pp 1–7

[34] Zou Y and Cheng L 2021 A transfer learning model for gesture recognition based on the deep features extracted by CNN *IEEE Trans. Artif. Intell.* **2** 447–58

[35] Jiang Y, Song L, Zhang J, Song Y and Yan M 2022 Multi-category gesture recognition modeling based on sEMG and IMU signals *Sensors* **22** 5855

[36] Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G and Gelly S 2020 An image is worth 16x16 words: transformers for image recognition at scale (arXiv:2010.11929)

[37] Herrmann C, Sargent K, Jiang L, Zabih R, Chang H, Liu C, Krishnan D and Sun D 2022 Pyramid adversarial training improves vit performance *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition* pp 13419–29

[38] Shin J, Musa Miah A S, Hasan M A M, Hirooka K, Suzuki K, Lee H-S and Jang S-W 2023 Korean sign language

recognition using transformer-based deep neural network *Appl. Sci.* **13** 3029

[39] Guo J, Han K, Wu H, Tang Y, Chen X, Wang Y and Xu C 2022 Cmt: convolutional neural networks meet vision transformers *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition* pp 12175–85

[40] Long F, Qiu Z, Pan Y, Yao T, Luo J and Mei T 2022 Stand-alone inter-frame attention in video models *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition* pp 3192–201

[41] Hoyso 2023 Isolated sign language recognition (available at: www.kaggle.com/competitions/asl-signs/discussion/406684)

[42] Devlin J, Chang M-W, Lee K and Toutanova K 2018 Bert: pre-training of deep bidirectional transformers for language understanding (arXiv:1810.04805)

[43] Chumachenko K, Iosifidis A and Gabbouj M 2022 Self-attention fusion for audiovisual emotion recognition with incomplete data *2022 26th Int. Conf. on Pattern Recognition (ICPR)* (IEEE) pp 2822–8

[44] Atzori M, Gijsberts A, Castellini C, Caputo B, Hager A G, Elsig S, Giatsidis G, Bassetto F and Muller H 2014 Electromyography data for non-invasive naturally-controlled robotic hand prostheses *Sci. Data* **1** 140053

[45] Cescon C, Farina D, Gobbo M, Merletti R and Orizio C 2004 Effect of accelerometer location on mechanomyogram variables during voluntary, constant-force contractions in three human muscles *Med. Biol. Eng. Comput.* **42** 121–7

[46] Smith L H, Hargrove L J, Lock B A and Kuiken T A 2011 Determining the optimal window length for pattern recognition-based myoelectric control: balancing the competing effects of classification error and controller delay *IEEE Trans. Neural Syst. Rehabil. Eng.* **19** 186–92

[47] Tigrini A, Al-Timemy A H, Verdini F, Fioretti S, Morettini M, Burattini L and Mengarelli A 2023 Decoding transient sEMG data for intent motion recognition in transhumeral amputees *Biomed. Signal Process. Control* **85** 104936

[48] Khushaba R N and Nazarpour K 2021 Decoding hd-emg signals for myoelectric control-how small can the analysis window size be? *IEEE Robot. Autom. Lett.* **6** 8569–74

[49] Gijsberts A, Atzori M, Castellini C, Muller H and Caputo B 2014 Movement error rate for evaluation of machine learning methods for sEMG-based hand movement classification *IEEE Trans. Neural Syst. Rehabil. Eng.* **22** 735–44

[50] Atzori M, Gijsberts A, Müller H and Caputo B 2014 Classification of hand movements in amputated subjects by sEMG and accelerometers *2014 36th Annual Int. Conf. IEEE Engineering in Medicine and Biology Society* (IEEE) pp 3545–9

[51] Montazerin M, Zabihi S, Rahimian E, Mohammadi A and Naderkhani F 2022 ViT-HGR: vision transformer-based hand gesture recognition from high density surface EMG signals *2022 44th Annual Int. Conf. IEEE Engineering in Medicine & Biology Society (EMBC)* (IEEE) pp 5115–9

[52] Xiao T, Singh M, Mintun E, Darrell T, Dollár P and Girshick R 2021 Early convolutions help transformers see better *Advances in Neural Information Processing Systems* vol 34 pp 30392–400

[53] Bangaru S S, Wang C, Busam S A and Aghazadeh F 2021 ANN-based automated scaffold builder activity recognition through wearable EMG and IMU sensors *Autom. Constr.* **126** 103653

[54] Boschmann A and Platzner M 2014 Towards robust HD EMG pattern recognition: reducing electrode displacement effect using structural similarity *2014 36th Annual Int. Conf. IEEE Engineering in Medicine and Biology Society* (IEEE) pp 4547–50

[55] Gerdle B, Karlsson S, Crenshaw A and Friden J 1997 The relationships between EMG and muscle morphology throughout sustained static knee extension at two submaximal force levels *Acta Physiol. Scand.* **160** 341–51

[56] Yang J, Liu J, Xu N and Huang J 2023 Tvt: transferable vision transformer for unsupervised domain adaptation *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision* pp 520–30

[57] Fatayer A, Gao W and Fu Y 2022 sEMG-based gesture recognition using deep learning from noisy labels *IEEE J. Biomed. Health Inf.* **26** 4462–73

[58] Zhai X, Jelfs B, Chan R H M and Tin C 2017 Self-recalibrating surface EMG pattern recognition for neuroprosthesis control based on convolutional neural network *Front. Neurosci.* **11** 266372